Related Rates

ANSWER KEY

Setting up related-rates problems

1 A circular oil spill is expanding so that its radius increases at 3 m/min. How fast is the area increasing when the radius is 50 m?

$A = \pi r^2$, so $\frac{dA}{dt} = 2\pi r\frac{dr}{dt} = 2\pi(50)(3) = 300\pi \approx 942.5$ m^2/min.

2 A 10 m ladder leans against a wall. The bottom slides away from the wall at 2 m/s. How fast is the top sliding down when the bottom is 6 m from the wall?

$x^2 + y^2 = 100$. Differentiating: $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$. At $x = 6$, $y = 8$: $12(2) + 16\frac{dy}{dt} = 0$, so $\frac{dy}{dt} = -1.5$ m/s (the top is sliding down at 1.5 m/s).

Volume and other rates

3 Air is pumped into a spherical balloon at 100 cm^3/s. Find the rate of change of the radius when $r = 5$ cm, to three decimal places.

$V = \frac{4}{3}\pi r^3$, so $\frac{dV}{dt} = 4\pi r^2\frac{dr}{dt}$. $100 = 4\pi(25)\frac{dr}{dt}$, so $\frac{dr}{dt} = \frac{1}{\pi} \approx 0.318$ cm/s.

4 Water drains from a conical tank (vertex down) with radius 4 m and height 10 m at the top, at a rate of 2 m^3/min. Using similar triangles $r = \frac{2h}{5}$, find how fast the water level is dropping when $h = 5$ m.

$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi\left(\frac{2h}{5}\right)^2 h = \frac{4\pi}{75}h^3$. So $\frac{dV}{dt} = \frac{12\pi}{75}h^2\frac{dh}{dt}$. At $h = 5$: $-2 = \frac{12\pi}{75}(25)\frac{dh}{dt}$, giving $\frac{dh}{dt} = -\frac{2 \times 75}{300\pi} = -\frac{1}{2\pi} \approx -0.159$ m/min.

Angle and distance rates

5 A camera tracks a rocket launching vertically, 500 m from the launch pad. When the rocket is 1200 m high and rising at 200 m/s, how fast is the camera's angle of elevation increasing? Give the answer in radians/s, to four decimal places.

$\tan\theta = \frac{h}{500}$, so $\sec^2\theta\frac{d\theta}{dt} = \frac{1}{500}\frac{dh}{dt}$. At $h = 1200$: hypotenuse $= \sqrt{500^2 + 1200^2} = 1300$, $\sec^2\theta = \left(\frac{1300}{500}\right)^2 = 6.76$. $6.76\frac{d\theta}{dt} = \frac{200}{500} = 0.4$, so $\frac{d\theta}{dt} \approx 0.0592$ rad/s.

6 Two cars leave an intersection at the same time, one heading north at 40 mph and the other heading east at 30 mph. How fast is the distance between them increasing after 2 hours?

$x = 60$ mi, $y = 80$ mi (east, north after 2 h). $s^2 = x^2 + y^2$: $2s\frac{ds}{dt} = 2x\frac{dx}{dt} + 2y\frac{dy}{dt}$. $s = 100$: $100\frac{ds}{dt} = 60(30) + 80(40) = 1800 + 3200 = 5000$, so $\frac{ds}{dt} = 50$ mph.

More related rates

7 A streetlight is 6 m high. A 1.8 m tall person walks away from it at 1.5 m/s. Using similar triangles, find how fast the length of the person's shadow is growing.

Let x = distance from pole, s = shadow length. Similar triangles: $\frac{6}{x+s} = \frac{1.8}{s}$, so $6s = 1.8x + 1.8s$, giving $4.2s = 1.8x$, $s = \frac{1.8}{4.2}x = \frac{3}{7}x$. $\frac{ds}{dt} = \frac{3}{7} \cdot 1.5 \approx 0.643$ m/s.

8 The volume of a cube is increasing at 12 cm^3/s. Find the rate of change of the side length when the side is 3 cm.

$V = s^3$, $\frac{dV}{dt} = 3s^2\frac{ds}{dt}$. $12 = 3(9)\frac{ds}{dt}$, so $\frac{ds}{dt} = \frac{12}{27} = \frac{4}{9}$ cm/s.

Rates involving trigonometric relationships

9 A kite is flying at a constant height of 40 m, moving horizontally away from the person holding the string at 3 m/s. Find the rate at which the string length is increasing when the kite is 30 m horizontally from the person.

$L^2 = 40^2 + x^2$: $2L\frac{dL}{dt} = 2x\frac{dx}{dt}$. At $x = 30$: $L = \sqrt{1600 + 900} = 50$. $50\frac{dL}{dt} = 30(3) = 90$, so $\frac{dL}{dt} = 1.8$ m/s.

10 A conical pile of sand has its height always equal to its base radius. Sand is added at 10 m^3/min. Find the rate of change of the height when $h = 5$ m, using $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi h^3$.

$\frac{dV}{dt} = \pi h^2\frac{dh}{dt}$. At $h = 5$: $10 = \pi(25)\frac{dh}{dt}$, so $\frac{dh}{dt} = \frac{10}{25\pi} = \frac{2}{5\pi} \approx 0.127$ m/min.

Visualizing a related-rates scenario

11 The graph shows $A(r) = \pi r^2$, the area of the oil spill from the first problem, as a function of radius alone (not time). Explain why $\frac{dA}{dt}$ still depends on $\frac{dr}{dt}$ even though this graph has no time axis.

The graph shows how A depends on r, but r itself changes with time. The chain rule $\frac{dA}{dt} = \frac{dA}{dr} \cdot \frac{dr}{dt} = 2\pi r \cdot \frac{dr}{dt}$ combines the graph's slope ($\frac{dA}{dr} = 2\pi r$) with the given rate $\frac{dr}{dt}$ to produce the actual rate of change over time.

Related rates with area and perimeter together

12 A rectangle's length is always twice its width. If the width is increasing at 3 cm/s, find the rate of change of the area when the width is 5 cm.

$A = w(2w) = 2w^2$. $\frac{dA}{dt} = 4w\frac{dw}{dt}$. At $w = 5$: $\frac{dA}{dt} = 4(5)(3) = 60$ cm^2/s.

13 For the same rectangle (length $= 2 \times$ width), find the rate of change of the perimeter when the width is increasing at 3 cm/s.

$P = 2(w + 2w) = 6w$. $\frac{dP}{dt} = 6\frac{dw}{dt} = 6(3) = 18$ cm/s (constant, independent of w, since P is linear in w).

A two-stage related-rates problem

14 A spherical balloon is being inflated so that its surface area $S = 4\pi r^2$ increases at 12π cm^2/s. Find $\frac{dr}{dt}$ when $r = 3$ cm, then use it to find $\frac{dV}{dt}$ at that instant.

$\frac{dS}{dt} = 8\pi r \frac{dr}{dt}$: $12\pi = 8\pi(3)\frac{dr}{dt}$, so $\frac{dr}{dt} = \frac{1}{2}$ cm/s. Then $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt} = 4\pi(9)\left(\frac{1}{2}\right) = 18\pi$ cm^3/s.